

BITS Pilani presentation

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ES ZG513/ SS ZG513, Network Security
Lecture No III.

Integers and Divisibility

Cryptography is built on properties of integers.

For integers a, b we say a divides b if

b = a*k for some integer k.

and write a | b.

Basic properties:

If a | b and a | c, then a | (bx+cy) for any integers x, y.

This property is crucial for GCD computation and modular inverses.

Prime Numbers

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A prime number has exactly two positive divisors:
1 and itself.

A composite number has more than two divisors.

Fundamental result:

Every integer
n>1 can be written uniquely as a product of primes
(Fundamental Theorem of Arithmetic).

Importance in cryptography:

RSA relies on large prime numbers. Security depends on the difficulty of prime factorization.

Modular Arithmetic

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For integers a and n>0:

a mod n = remainder when a is divided by n

Congruence relation:

$A \equiv b \pmod{n} \Leftrightarrow n \mid (a-b)$

Properties:

Addition, subtraction, and multiplication are well defined modulo n.

Division is not always possible modulo n.

Used in: Caesar cipher, Affine cipher, RSA, Diffie–Hellman, ECC.

Greatest Common Divisor (GCD)

The GCD of integers a and b, denoted gcd(a,b), is the largest integer dividing both.

If gcd(a,b)=1, then a and b are coprime.

Cryptographic importance:

An integer a has a multiplicative inverse modulo n iff gcd(a,n)=1

Euclidean Algorithm

Efficient algorithm to compute gcd(a,b).
Based on repeated division:
$a = bq + r, 0 \leq r < b$
Replace (a,b) by (b,r) and repeat until r=0.

Last non-zero remainder is the gcd.

Runs in polynomial time, even for very large numbers.

Extended Euclidean Algorithm

Extends Euclidean Algorithm to find integers x, y such that: $ax + by = gcd(a, b)$

If $gcd(a, n) = 1$, then:

$ax \equiv 1 \pmod{n}$

So x is the modular inverse of a.

Used in:

RSA key generation

Affine cipher

Hill cipher (matrix inverse modulo n)

Modular Inverses

An integer a has an inverse modulo n if: gcd(a,n)=1

The inverse $a^{-1} \equiv 1 \pmod{n}$

Essential for:

Decryption in affine cipher
Matrix inversion in Hill cipher
RSA private key computation

Euler's Totient Function

$\varphi(n)$: number of integers in $\{1,\ldots,n\}$ co-prime to n.

If $n=p$ is prime, $\varphi(p)=p-1$

If $n=pq$ where p,q are distinct primes,
$\varphi(n)=(p-1)(q-1)$

Used in:

Euler's theorem
RSA encryption and decryption

Euler's Theorem

If gcd(a,n)=1, then:
$a^{\varphi(n)} \equiv 1 \pmod{n}$

Generalizes Fermat's Little Theorem.

Mathematical foundation of RSA correctness.

Matrices and Notation

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A matrix is a rectangular array of numbers arranged in rows and columns.

An m×n matrix has m rows and n columns.

In cryptography, matrix entries are usually taken modulo n (typically n=26 or n=256).

Matrix Operations (Modulo n)

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(a) Matrix Addition

For matrices A and B:

$C=A+B(\mathrm{mod}n)$

Addition is done entry-wise. Used rarely in isolation, but important in linear transformations.

(b) Matrix Multiplication

For matrices A (m×k) and B (k×n):

$C=AB(\mathrm{mod}n)$

Defined only when inner dimensions match.

Order matters: $AB \neq BA$ in general.

Provides strong diffusion in cryptographic systems.

Representing Text as Vectors

Letters are mapped to numbers, e.g.:

A=0,B=1,…,Z=25

Plaintext is divided into blocks of fixed size.

Each block is treated as a column vector.

Example:

ACT

0

2

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Hill Cipher (Main Application)

Uses a key matrix K

Encryption: C=KP(mod26)

Decryption: P=K^{−1}C(mod26)

Determinant of a Matrix

The determinant is a scalar value associated with a square matrix.

A matrix is invertible iff its determinant is non-zero.

In modular arithmetic: gcd(det(K),n)=1

must hold for the inverse to exist.

Cryptographic importance:

If determinant is not invertible modulo n, decryption is impossible.

Matrix Inverse Modulo n

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For a matrix K:

$K^{-1}=(\det(K))^{-1}\cdot adj(K) \pmod n$

Steps:

Compute det(K).

Find $(\det(K))^{-1} \bmod n$ using Extended Euclidean Algorithm

Compute adjugate matrix

Multiply and reduce modulo n

Role in Modern Cryptography

Direct matrix ciphers are rarely used today.

Matrix operations are still fundamental in:

Linear transformations

Block ciphers (internally)

Lattice-based cryptography (advanced topic)